

Classification of Elements and Periodicity in Properties

SHORT NOTES

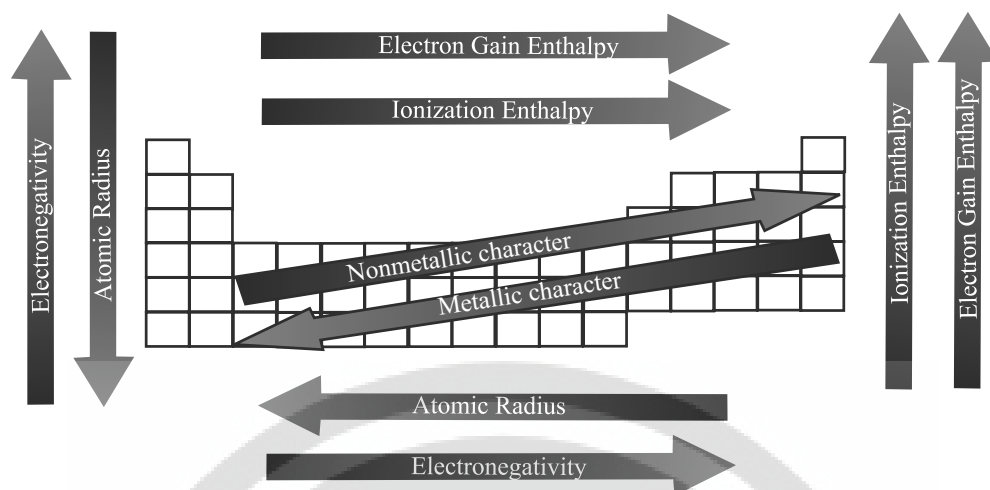


Fig. 10: The periodic trends of elements in the periodic table

Table: Periodic Trends in Valence of Elements as shown by the Formulas of Their Compounds

Group	1	2	13	14	15	16	17	18
Number of valence electron	1	2	3	4	5	6	7	8
Value	1	2	3	4	3.5	2.6	1.7	0.8

Group	1	2	13	14	15	16	17
Formula of hydride	LiH NaH KH	CaH ₂	B ₂ H ₆ AlH ₃	CH ₄ SiH ₄ GeH ₄ SnH ₄	NH ₃ PH ₃ AsH ₃ SbH ₃	H ₂ O H ₂ S H ₂ Se H ₂ Te	HF HCl HBr HI
Formula of oxide	Li ₂ O Na ₂ O K ₂ O	MgO CaO SrO BaO	B ₂ O ₃ Al ₂ O ₃ Ga ₂ O ₃ In ₂ O ₃	CO ₂ SiO ₂ GeO ₂ SnO ₂ PbO ₂	N ₂ O ₃ , N ₂ O ₅ P ₄ O ₆ , P ₄ O ₁₀ As ₂ O ₃ , As ₂ O ₅ Sb ₂ O ₃ , Sb ₂ O ₅ Bi ₂ O ₃ –	SO ₃ SeO ₃ TeO ₃ –	– Cl ₂ O ₇ – –

Mnemonic

1	2	13	14	15	16	17	18
H Hey							He honey
Li Li	Be beta	B baingan	C khe	N nana	O old	F fir	Ne ne
Na Na	Mg maange	Al aalu	Si sita	P patikar	S style	Cl kal	Ar aa
K ki	Ca car	Ga gaajar	Ge gi	As aishwarya	Se se	Br baher	Kr kar
Rb rab	Sr scooter	In in	Sn suno	Sb sb	Te te	I aayi	Xe xenon
Cs se	Ba baap	Tl thela	Pb prabhu	Bi bimar	Po po	At anty	Rn rakha
Fr fariyad	Ra raazi						

Scary	Tiny	Vicious	Creatures	May	Fear	Cows	[&]	Nice	Cute
21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn

Yes	Zebras	Never (But)	Most	Technicians	Run	Rhymes	Purely	Sweet	[&]	Cute
39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	